Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/21/2025
PI Name:	Britten, Richard Ph.D.		
Project Title:	Changes in the Neuroproteome Associated with HZE-Induced Impairment of Cognition		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	02/27/2014	**End Date:**	10/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Element change from SR to HFBP per Human Research Roadmap information dtd July 2019 (Ed., 1/6/2020) NOTE: End date changed to 10/31/2025 per S. Mack-Phillips / NASA JSC (Ed., 6/2/25) NOTE: Extended to 6/30/2025 per S. Mack-Phillips / NASA JSC (Ed., 5/16/24) NOTE: Extended to 6/30/2024 per NSSC information (Ed., 8/3/23) NOTE: Extended to 4/30/2023 per NSSC information (Ed., 12/30/2020) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 3/12/19) NOTE: Extended to 12/31/2018 per S. Monk/SR/LaRC (Ed., 1/11/18)		

Key Personnel Changes/Previous PI:	April 2025 update per the PI: Due to the late stage of the project, Co-Investigators are no longer with the project. These have been removed from the list of Co-Investigators for this reporting period: Douglas Wallace, Ashley Blackwell, Biswajit Maharathi (Ed., 4/17/25). December 2022 report: Dr. Dorela Shuboni-Mulligan of Eastern Virginia Medical School has been added to the project to determine if the SR-induced cognitive decrements are related to changes in the retina, specifically the loss of intrinsically photosensitive retinal ganglion cells (ipRGCs). December 2022 report: Dr. Biswajit Maharathi from the Illinois Brain Analytics Institute (University of Illinois Chicago) has been added to the project to assist in applying AI-assisted learning algorithms to determine neural network cohesiveness. December 2021 report: Dr. Jessica Burket at Christopher Newport University has been added to the project to conduct the social interaction studies. Dr Ashley Blackwell at Eastern Virginia Medical School has been added to the project to conduct the string pulling and neural network cohesiveness studies. December 2019 report: Dr. Douglas Wallace at Northern Illinois University (NIU) is now a CoInvestigator. Dec 2016: Drs. Semmes and Dutta were removed from the project; proteomic analysis is now being conducted at UTMB (University of Texas Medical Branch) as contract work.
COI Name (Institution):	Burket, Jessica Ph.D. (Christopher Newport University) Shuboni-Mulligan, Dorela (Eastern Virginia Medical School)
Grant/Contract No.:	NNX14AE73G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that High Z, High Energy (HZE) particles have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks (spatial memory and Executive function-attentional set shifting) following exposure to low HZE doses. The goal of this application is to determine the Threshold dose for the induction of HZE-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) following exposure to 56Fe, 48Ti, and 28Si particles. The proposed studies will also identify the changes in the proteome of the brain (neuroproteome) of rats that differ in their susceptibility to HISMI and HIASSI, which will provide further insight into the factors that lead to HISMI/HIASSI and perhaps more importantly, that prevent its emergence. Our underlying hypothesis is that HISMI and HIASSI arise as the direct result of HZE-induced changes in the neuroproteome. We also hypothesize that exposure to HZE species that have different track structures will result in different mechanisms of HZE-induced cognitive impairment (HICI). Collectively, these studies will give some insight into the underlying cause for HISMI and HIASSI. Our studies will thus address CNS (Central Nervous System) Gaps 1, 2, and 6, [Ed. note February 2022: Human Research Program risks and gaps have changed since this project was initiated--see Human Research Roadmap for updated gaps: https://] and we shall specifically focus on the following aims: Aim 1. Determine the Threshold dose for the induction of HISMI and HIASSI following exposure to 56Fe, 48Ti, and 28Si particles when delivered as a single dose. Aim 2. Identify changes in the neuroproteome that are associated with susceptibility or resistance to developing HISMI and HIASSI following exposure to 56Fe particles. Aim 3. Determine the mechanism of HISMI and HIASSI induced by HZE particles of differing LET (linear energy transfer). In Aim 1, socially mature (~6 month old) male Wistar rats will be irradiated with 56Fe, 48Ti, and 28Si particles (with incident energies of 600 MeV/nucleon). Rats will receive whole body HZE irradiation (< 15 cGy), and HISMI and HIASSI will be assessed at 3 months post irradiation. In Aim 2 and 3, the composition of the neuroproteome (hippocampus and selected regions of the prefrontal cortex) of irradiated rats that have “normal” cognitive performance or have developed HISMI or HIASSI will be established using an unbiased proteomic profiling approach. We shall use a label free differential protein profiling workflow on the Q-Exactive Orbitrap mass spectrometer. These studies will give considerable insight into the underlying cause for HZE-induced neurocognitive failure. The proposed studies will continue to define the minimum dose of HZE particles that will induce HISMI and HIASSI. Moreover, our studies will provide considerable insight into the underlying mechanism of HICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of HICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA’s medical staff can implement appropriate countermeasures to protect these at risk individuals. Supplemental studies (in December 2019 report) This study will provide information on the robustness of single-exposure experiments to predict the impact of repeated episodic radiation exposures (such as will be encountered on the mission to Mars) on neurocognition. This study will test the hypothesis that episodic SR exposure will result in more severe neurocognitive deficits than single, or multiple daily SR doses. In addition, this study will be a robust (akin to a pHase III clinical trial) concurrent validation of the effect of a single dose of 10 cGy simplified 6-ion GCRSim versus a single dose of 10 cGy 250 MeV/n He ions on ATSET/UCFlex performance using the same batch of rats, laboratory personal, transport and environmental conditions. This study will utilize both male and female rats, and two different radiation regimens incorporating 4He ions and the 6-ion GCRsim beam. Executive function performance (ATSET) will be assessed after a single exposure (He or GCRsim) and after a second exposure (~6 months later) to the 6-ion GCRsim beam. To maximize the amount of data obtained from these expensive studies, where possible (dependent upon volunteers in the Britten lab) the impact of these radiations on sensorimotor (string pulling activity), social interaction and switch task performance will also be established.
Rationale for HRP Directed Research:	

Research Impact/Earth Benefits:

These studies will give considerable insight into the underlying cause for Space radiation (SR)-induced neurocognitive impairment (SICI).
The proposed studies will continue to define the minimum dose of SR particles that will impair cognitive flexibility (Attentional Set shifting and Unconstrained cognitive flexibility) performance. Importantly both of our cognitive flexibility tasks are homologs of tasks used in clinical testing of humans. Our studies will model the impact that single and repeated episodic exposure to SR has on neurocognitive performance and fine motor skills.

Moreover, our studies will provide considerable insight into the underlying mechanism of SICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of SICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA's medical staff can implement appropriate countermeasures to protect these at risk individuals.

Task Progress:

Project Objectives
1. Identify the lowest dose of space radiation (SR) that results in Attentional Set Shifting (ATSET) impairment. 2. Determine if there are LET-specific mechanisms of ATSET impairment. 3. Identify changes in the neuroproteome that reflect the cognitive performance status of SR-exposed animals.

Supplemental Studies

4. Establish the impact that re-irradiation with 10 cGy of simplified (5-ion) CGRsim beam has on the ATSET performance of male and female Wistar rats that maintained a functional ATSET performance after exposure to 10 cGy of either He or GCRsim. 5. Establish the impact that exposure to 10 cGy of either He or GCRSim has on the risk-taking predisposition (RTP) of male and female rats.

Project Approach

To better simulate the "clinical reality," adult rats that have been pre-selected for good ATSET performance, and who have been maintained on an exercise regimen are used in this study. The first radiation exposures will occur when the rats are ~7 months old. While the biological equivalent age of these rats is closer to that of a 30-year-old human, which is currently younger than most astronauts, the use of such rats allows for the long-term monitoring of cognitive decline, which is less likely to be impacted by age-related cognitive decline.

Rats are exposed to 10 cGy of SR ions and ATSET performance re-established at 3 months post-exposure. After completion of the ATSET test, the rats are then tested in the Unconstrained Cognitive flexibility (UCFlex) assay, which requires the rats to develop a novel solution to obtain the food reward. Thus, the UCFlex version of the ATSET task interrogates both constrained and unconstrained cognitive flexibility performance within individual rats. Importantly, both cognitive flexibility tasks are homologs of tasks used in clinical testing of humans.

Most of our studies have established the impact that a single dose of SR has on cognitive performance. However, this widely used approach may not be the optimal one for risk estimation studies for long-duration space missions, as the derived risk estimates are likely to be pertinent only for the early stages of a deep space mission, and even then only for astronauts that have no previous experience in space. Given the multitude of SR-induced CNS changes observed at the structural, neurotransmitter and protein levels, previously exposed brains may have a markedly different response to SR exposure than that seen in naïve brains. The supplemental studies were awarded to determine how prior SR exposure alters the severity, nature or frequency of ATSET impairment compared to that observed in radiation-naïve individuals. These studies involve returning the rats that have maintained good ATSET performance after SR (10 cGy 4He ions or the 5-ion GCRsim beam) to BNL where they receive a second dose of 10 cGy n GCRsim. Cognitive performance is then reassessed at 3 months after the second exposure.

Executive functions also regulate psychological processes, such as impulsivity, risk-taking, mood and social interactions; should SR-exposure alter these executive functions as it does cognitive flexibility, there is the possibility of altered inter-crew interactions and team cooperativity during prolonged space exploration. We have previously reported that exposure to 5 cGy He ions leads to social withdrawal (within freely interacting dyads) in male Wistar rats (Burket et al., 2021). Supplemental funding (an HRP-IWS student award) has allowed us to determine if SR exposure alters the Risk Taking Predisposition (RTP) (gambling/poor decision-making) of rats. We have determined that in both male and female rats that were preselected for inherently low RTP, SR resulted in altered RTP. At 30 days post-SR exposure, female rats showed an increased tendency to make poor decisions, while male rats retained their original RTP tendencies, but took longer to make a decision. A similar phenomenon was observed in rats that had had no prior exposure to the RTP task. SR exposure led to an increased RTP in both male and female rats.

The current reporting period saw the completion of our studies on the effects of SR on casual reasoning under high and low cognitive stress (multi-tasking). We have confirmed that both male and female rats have impaired performance in the ARMIT task, which is consistent with an inability to suppress the impact of irrelevant cues to complete the task. These data strongly suggest that the Central Executive Control neural network may be compromised following SR exposure.

These findings are now being investigated by establishing changes in the activity of various brain regions involved in that network. These studies utilize a state-of-the-art technique (developed in part using NASA funds) that allows for the activity of up to 4 brain regions to be determined (in real-time) as rats are actively engaged in a task. This technique essentially mimics studies in humans that use fMRI to identify neural network activity.

Research Highlights from this reporting period. The completion of many studies has enabled several "general" conclusions to be made:

• Low doses (10 cGy) of SR lead to the loss of performance in multiple executive functions, including the ATSET, UCFlex, Switch, RTP, and ARMIT tasks. • In most cases, both male and female rats have impaired executive function performance; however, the exact processes within a task may vary, as well as the type of performance decrements. There can be sex-dependent differences in the ability of rats to complete a task or the speed at which they can complete the task (processing) speed. • An assessment of performance savings (a concept widely used in many fields to define the faster response to a situation that has been previously encountered than when it was initially encountered) revealed that SR exposure eliminated performance savings in the ATSET task. This was observed in both males and females, but the stage that was impacted differed in a sex-dependent manner. The loss of performance savings was observed after both

	single and double SR-exposures. • A double exposure to SR leads to additional performance decrements; thus, the concept that there are intrinsically radioresistant rats appears to be NOT valid. • The type of performance decrements observed in the doubly-irradiated differs from that observed after a single exposure, often being manifested in a different stage and in a different manner (processing speed). There are sex-dependent differences in the performance decrements. • Our data suggests that risk estimates derived from studies that use radiation naïve rats may not fully reflect the incidence and nature of ATSET performance deficits that could occur over the entire duration of a mission to Mars, or in astronauts who return to deep space on multiple occasions. • The ability to monitor neural network activity in rats when under cognitive loading provides an opportunity to more extensively identify how SR impacts the dynamic interaction between brain regions.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Britten RA, Fesshaye AS, Tidmore A, Tamgue EN, Alvarado-Arriaga PA. "Different spectrum of space radiation induced cognitive impairments in radiation-naïve and adapted rats." Life Sci Space Res. 2024 Nov;43:68-74. https://doi.org/10.1016/j.lssr.2024.09.001 ; PubMed PMID: 39521496 , Nov-2024
Articles in Peer-reviewed Journals	Li N, Phuyal S, Smits E, Reid S, Tamgue EN, Arriaga PA. Britten RA. "Exposure to low (10cGy) doses of 4He ions leads to an apparent increase in risk taking propensity in female rats. " Behav Brain Res. 2024 Oct 2;474:115182. https://doi.org/10.1016/j.bbr.2024.115182 ; PubMed PMID: 39117150 , Oct-2024
Articles in Peer-reviewed Journals	Chang HHV, Fesshaye AS, Tidmore A, Sanford LD, Britten RA. "Sleep fragmentation results in novel set-shifting decrements in GCR-exposed male and female rats." Radiat Res. 2024 Nov 22;203(1):18-25. https://doi.org/10.1667/RADE-24-00146.1 ; PMID: 39576067 , Nov-2024
Articles in Peer-reviewed Journals	Smits E, Reid FE, Tamgue EN, Alvarado Arriaga P, Nguyen C, Britten RA. "Sex-dependent changes in risk-taking predisposition of rats following space radiation exposure." Life. 2025 Mar 12;15(3):449. https://www.mdpi.com/2075-1729/15/3/449 ; PMID: 40141792; PMCID: PMC11943666 , Mar-2025